Science A–Z ASSESSMENT

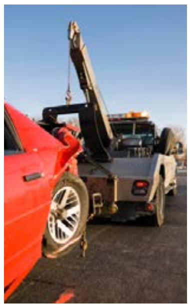
FORCE AND MOTION

Book Quiz

Name ______________________________ Date ____________

Directions: Read each question and choose the best answer.

1. For each motion shown below, identify the type of force required as either a *push* or a *pull*.

 Ⓐ ________ typing on a computer keyboard	 Ⓑ ________ removing the bow from the top of a present	 Ⓒ ________ dragging a backpack full of books
 Ⓓ ________ patting a friend on the back	 Ⓔ ________ wind knocking over an old tree	 Ⓕ ________ a truck towing a broken-down car

2. Which statement about force and motion is true?

   Ⓐ For every force, there is a motion.

   Ⓑ For every motion, there is a force.

   Ⓒ Force and motion have the same meaning.

   Ⓓ People and nature produce forces, while machines produce motion.

3. What will an object do if it has **inertia**?

   Ⓐ It will stay at rest until a force acts upon it.

   Ⓑ It will keep moving until a force acts upon it.

   Ⓒ It will keep moving or stay at rest forever.

   Ⓓ It will keep moving or stay at rest until a force acts upon it.

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4. What are Newton's laws of motion?

Ⓐ rules that people must follow when driving a car

Ⓑ rules that describe how energy moves from one object to another

Ⓒ rules that describe how all objects move

Ⓓ rules that explain how energy changes from one kind to another

5. If a friend bumps into the side of you while you are skating in a straight line, what will happen to your motion?

Ⓐ You will stop.

Ⓑ You will speed up.

Ⓒ You will keep skating in a straight line.

Ⓓ You will change direction.

6. According to Newton's second law of motion, which object will require the most force in order to change its motion?

Ⓐ the one with the greatest mass and fastest speed

Ⓑ the one with the greatest mass and slowest speed

Ⓒ the one with the least mass and slowest speed

Ⓓ the one with the least mass and greatest speed

7. Complete this sentence describing Newton's third law of motion:
For every action in one direction,

______________________________.

Ⓐ *a weaker force acts in the same direction*

Ⓑ *an equal force acts in the opposite direction*

Ⓒ *a stronger force acts in the opposite direction*

Ⓓ *an equal force acts in the same direction*

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8. A plant soaks up energy from the Sun and turns it into food so it can grow. A cow eats the plant so it can have energy to walk. You eat or drink products from the cow so you can have energy to play. You hit a ball, which makes a loud sound and sends the ball flying. Throughout these steps, the energy keeps moving and changing. What is this series of steps called?

Ⓐ energy movement
Ⓑ energy switching
Ⓒ energy transfer
Ⓓ energy conversion

9. Which task would require the most work?

Ⓐ carrying a 1 kilogram (2.2 lb.) bucket of water 3 meters (10 ft.)
Ⓑ carrying a 5 kilogram (11 lb.) bucket of water 7 meters (23 ft.)
Ⓒ carrying a 1 kilogram (2.2 lb.) bucket of water 7 meters (23 ft.)
Ⓓ carrying a 5 kilogram (11 lb.) bucket of water 3 meters (10 ft.)

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10. If you pull a rubber band back but don't release it, what kind of energy does the rubber band have?

Ⓐ kinetic energy

Ⓑ potential energy

Ⓒ wasted energy

Ⓓ motion energy

11. Rubbing your hands together is an example of *sliding friction,* while a paper plane flying through the air is an example of ____________.

Ⓐ *fluid friction*

Ⓑ *rolling friction*

Ⓒ *solid friction*

Ⓓ *heating friction*

12. How are the forces of magnetism and electricity related?

Ⓐ Magnets are made of metal, and metal is electric.

Ⓑ Both forces attract certain kinds of metals.

Ⓒ Both forces are only produced by people, not found in nature.

Ⓓ Magnets can produce electricity, and electricity can turn some metals into magnets.

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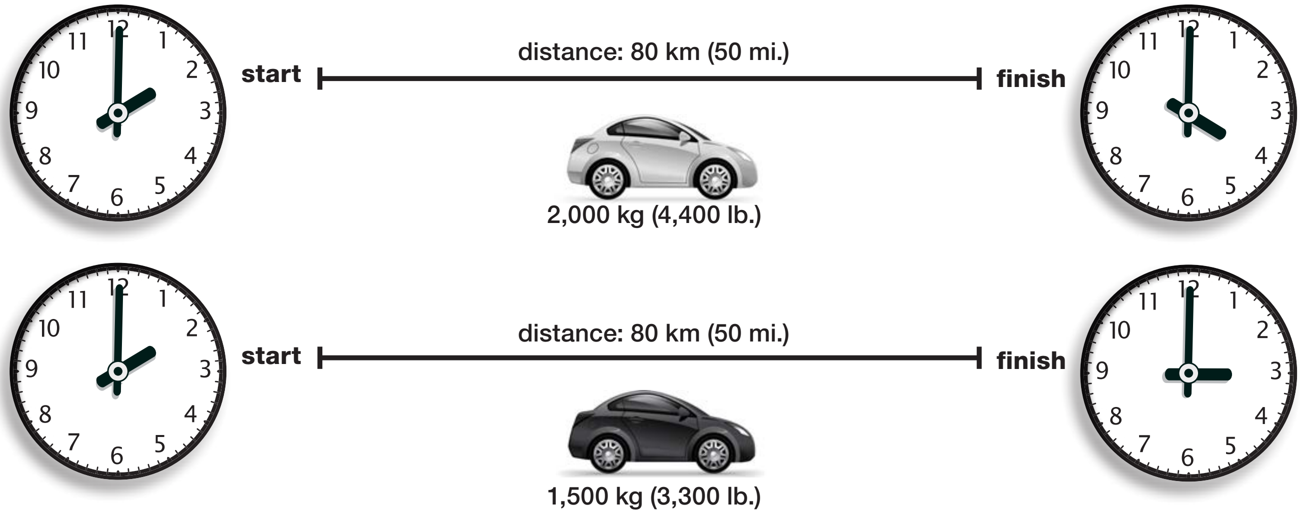
Directions: Use the diagram below to answer questions 13 and 14.



13. Which statement about the pull of gravity on the two cars is true?

Ⓐ The pull of gravity is stronger on the white car because it has greater mass.

Ⓑ The pull of gravity is stronger on the black car because it has greater mass.

Ⓒ The pull of gravity is the same on both cars because Earth's mass is so much greater than the mass of either car.

Ⓓ There is no pull of gravity because both cars are moving forward.

14. What is the speed of each car?

Ⓐ Both cars: 40 km/hr (25 mph)

Ⓑ Both cars: 80 km/hr (50 mph)

Ⓒ White car: 40 km/hr (25 mph); Black car: 80 km/hr (50 mph)

Ⓓ White car: 80 km/hr (50 mph); Black car: 40 km/hr (25 mph)

15. Extended Response: Think of something that moves. First, describe its motion, thinking about its speed, direction, or pattern. Then identify the force that causes the motion. Finally, explain the source of energy that provides the force.